Einstein and his collaborators fully recognized that they could not identify both the location and the velocity of any given particle. But, and this is key, even without determining both the position and velocity of either particle, EPR's reasoning shows that each *has* a definite position and velocity. To them, it was a question of reality. To them, a theory could not claim to be complete if there were elements of reality that it could not describe.

After a bit of intellectual scurrying in response to this unexpected observation, the defenders of quantum mechanics settled down to their usual, pragmatic approach, summarized well by the eminent physicist Wolfgang Pauli: "One should no more rack one's brain about the problem of whether something one cannot know anything about exists all the same, than about the ancient question of how many angels are able to sit on the point of a needle."[9] Physics in general, and quantum mechanics in particular, can deal only with the measurable properties of the universe. Anything else is simply not in the domain of physics. If you can't measure both the position and the velocity of a particle, then there is no sense in talking about whether it has both a position and a velocity.

EPR disagreed. Reality, they maintained, was more than the readings on detectors; it was more than the sum total of all observations at a given moment. When no one, absolutely no one, no device, no equipment, no anything at all is "looking" at the moon, they believed, the moon was still there. They believed that it was still part of reality.

In a way, this standoff echoes the debate between Newton and Leibniz about the reality of space. Can something be considered real if we can't actually touch it or see it or in some way measure it? In Chapter 2, I described how Newton's bucket changed the character of the space debate, suddenly suggesting that an influence of space could be observed directly, in the curved surface of spinning water. In 1964, in a single stunning stroke that one commentator has called "the most profound discovery of science,"[10] the Irish physicist John Bell did the same for the quantum reality debate.

In the following four sections, we will describe Bell's discovery, judiciously steering clear of all but a minimum of technicalities. All the same, even though the discussion uses reasoning less sophisticated than